Name: _____

Instructions: All solutions should be prepared carefully and neatly. Any questions that ask you to explain or justify should have answers written in complete sentences. All solution sets shall be completed on this packet and handed in at the beginning of class on the due date below. Careless presentation (e.g. bad handwriting, pen scribbles, doodles, wasted space, etc) will result in a deduction of points at my discretion. If the question asks to submit a graph or plot, be sure to print the plot and include it in your final lab report. Use a PENCIL and if you make a mistake, use an eraser. Each student should prepare their own solutions that demonstrate their own comprehension of the material. Submitted work that does not demonstrate clearly the process by which one arrived at the answer will not receive credit of any kind. Academic dishonesty will not be tolerated.

Desmos Lab IV

MATH 181 – Calculus I

Due: Friday, December 5

Read: Section 5.3 and 5.5

Problem Number	Available Points	Your Points
1	10	
2	10	
Total	20	

1. Definite Integrals: In this part, we'll use Desmos to compute definite integrals and validate the constant multiple property of integrals.
   (a) Open a new graph in Desmos and save it as "Definite Integrals."
   (b) Define the following function on the first line in Desmos: $f(x) = 4x^3 - 2x + 2$. Define the domain of the function from -1to1 by typing $\{-1 \leq x \leq 1\}$ right next to the function. Graph the function on the window $-2 \leq x \leq 2$ and $-0.5 \leq y \leq 5$.
   (c) On the next line, type $0 \leq y \leq f(x)\,\{-1 \leq x \leq 1\}$. What does Desmos do? Explain briefly below.

   (d) Compute the integral $\int_{-1}^{1} 4x^3 - 2x + 2\,dx$ by hand in the space below using the Fundamental Theorem of Calculus.

   (e) Now, on a new line, use Desmos to check your answer by typing in the integral exactly how it looks above.
   (f) Graph the area below $y = 2f(x)$ by typing $0 \leq y \leq 2f(x)\,\{-1 \leq x \leq 1\}$. Have Desmos calculate this area. Is the answer what you expect? Explain briefly.

   (g) When you submit this assignment on D2L, share the **link** to this graph

2. Symmetric Functions and Area: In this part, we'll use Desmos to explore how areas bounded by odd and even functions result in a somewhat easier calculation.
   (a) Open a new graph in Desmos and save it as "Symmetric Functions."
   (b) Create a folder called "Even Functions." Define the following function on the first line in the folder: $f_e(x) = -3x^4 + 4x^2 + 2$. Define the domain of the function from $-a$ to a by typing $\{-a \leq x \leq a\}$ right next to the function. Define a slider for a and let a range from 0 to 5.
   (c) On the next two lines in the folder, type $0 \leq y \leq f_e(x)\,\{-a \leq x \leq a\}$ and $f_e(x) \leq y \leq 0\,\{-a \leq x \leq a\}$. Push play on a. What does Desmos do? Explain briefly below.

   (d) Outside the folder, have Desmos calculate the following integral: $\int_{-a}^{a} f_e(x), dx$. Then on a new line, have it calculate $\int_{0}^{a} f_e(x), dx$. What is the relationship between these two areas? Explain.

   (e) Toggle off the graph of the even function by clicking the folder. Now, create a new folder called "Odd Functions." Define an odd function of your choice in this folder and call it $f_o(x)$. Plot the area between $-a$ and a, just like you did for the even function. Now, have Desmos calculate the area $\int_{-a}^{a} f_o(x), dx$. Is the area independent of a? Why? Explain briefly.

   (f) When you submit this assignment on D2L, share the **link** to this graph